

Comptia A+ Assignment

[A+ - Troubleshooting and Helpdesk]

Tools for Troubleshooting.

Overview of tools (multimeter, POST card, software diagnostics).

Task

❖ 1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

- A multimeter is an electronic measuring instrument used to measure voltage (V), current (A), resistance (Ω), and continuity. It helps technicians identify electrical faults in computers, smartphones, and other electronic devices.

Scenario	How the Multimeter Helps
Desktop PC does not power on.	The multimeter checks the <u>SMPS power output</u> (3.3V, 5V, and 12V) to determine whether the power supply is working correctly.
Smartphone is not charging.	The multimeter measures the <u>charger and battery voltage</u> to identify whether the charger, charging cable, battery, or charging port is faulty.

Laptop suddenly shuts down or does not start.	The multimeter checks the <u>laptop adapter output voltage</u> and the DC power jack to confirm that the correct power is reaching the motherboard.
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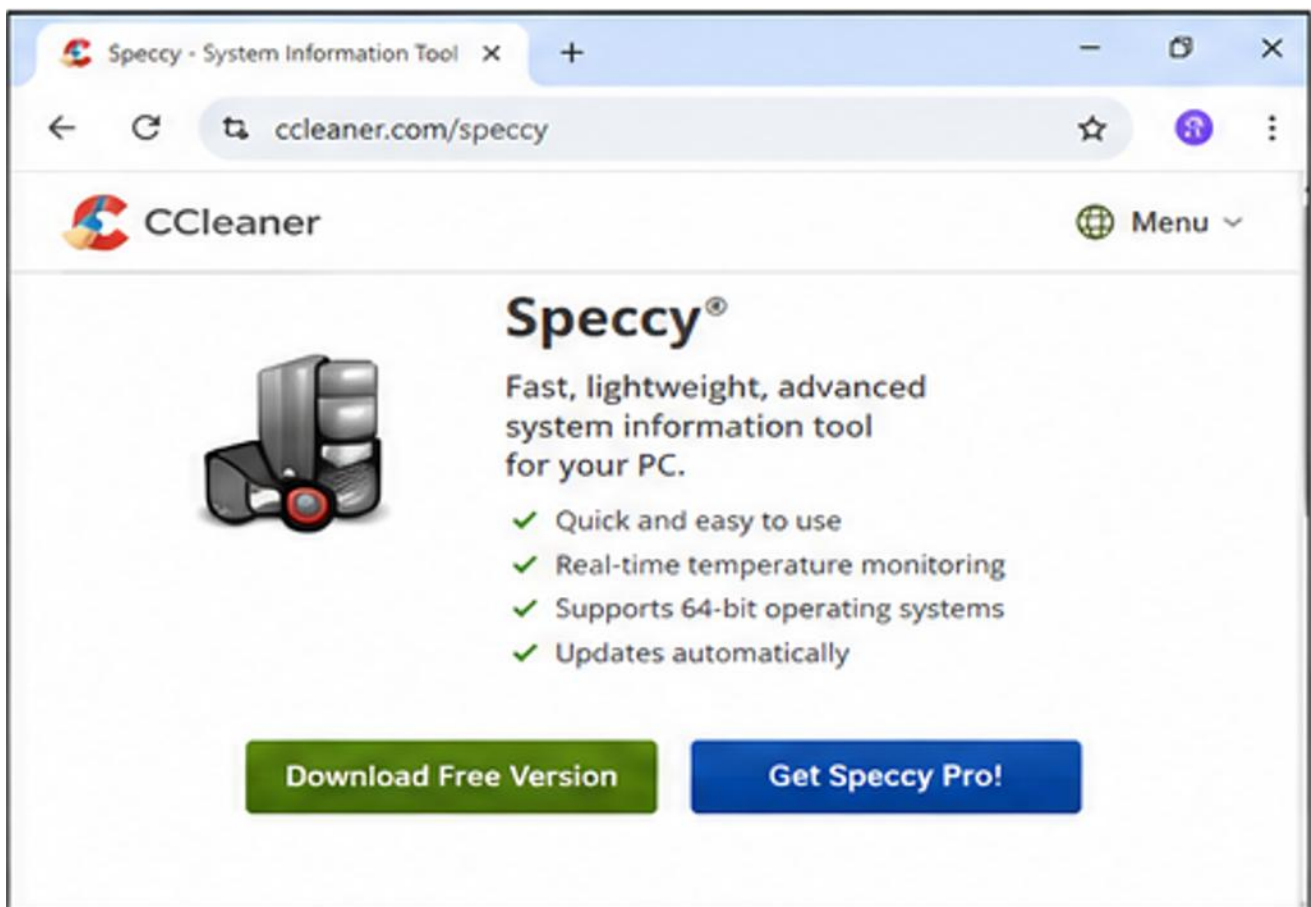
❖ **2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Focus on the codes shown and how they relate to hardware issues.**

➤ Summary:

➤ I watched a demonstration of a POST (Power-On Self-Test) card used for motherboard troubleshooting. The POST card displayed hexadecimal BIOS POST codes that changed as the motherboard tested different hardware components during startup. In the demonstration, the final POST code helped identify that the boot process had stopped because of a hardware fault, allowing the technician to narrow down the issue without replacing unnecessary parts. POST cards are useful for diagnosing motherboard, RAM, CPU, and BIOS startup problems.

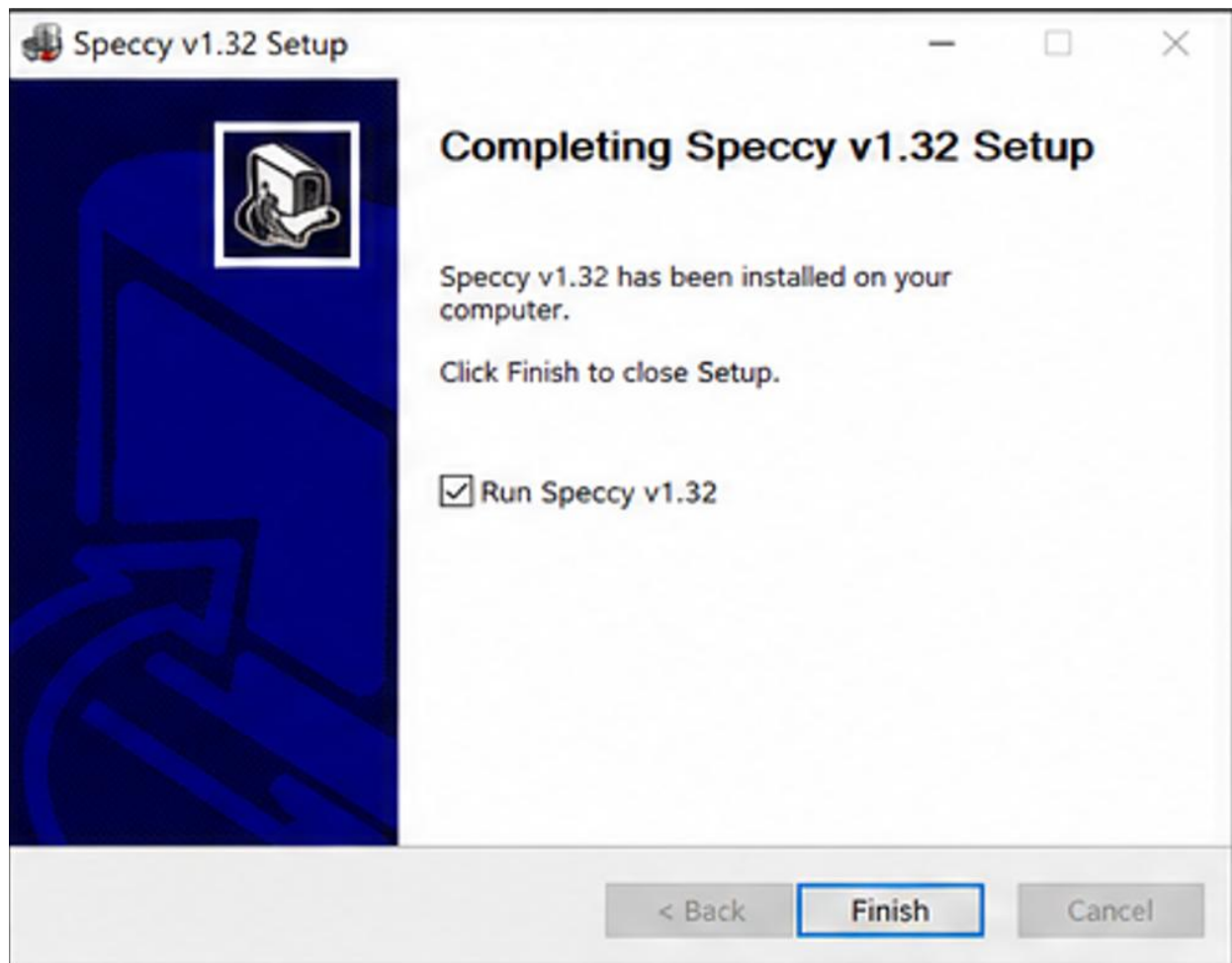
❖ **3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Blur or crop out any personal information before submitting.**

- Software Used Speccy (Free System Information & Diagnostic Tool:
- Step 1: Download the Software:
 - Visit the official Speccy website and download the latest version of the software.



➤ Step 2: Install Speccy:

- Run the downloaded setup file and complete the installation by following the on-screen instructions.



➤ Step 3: Launch the Application:

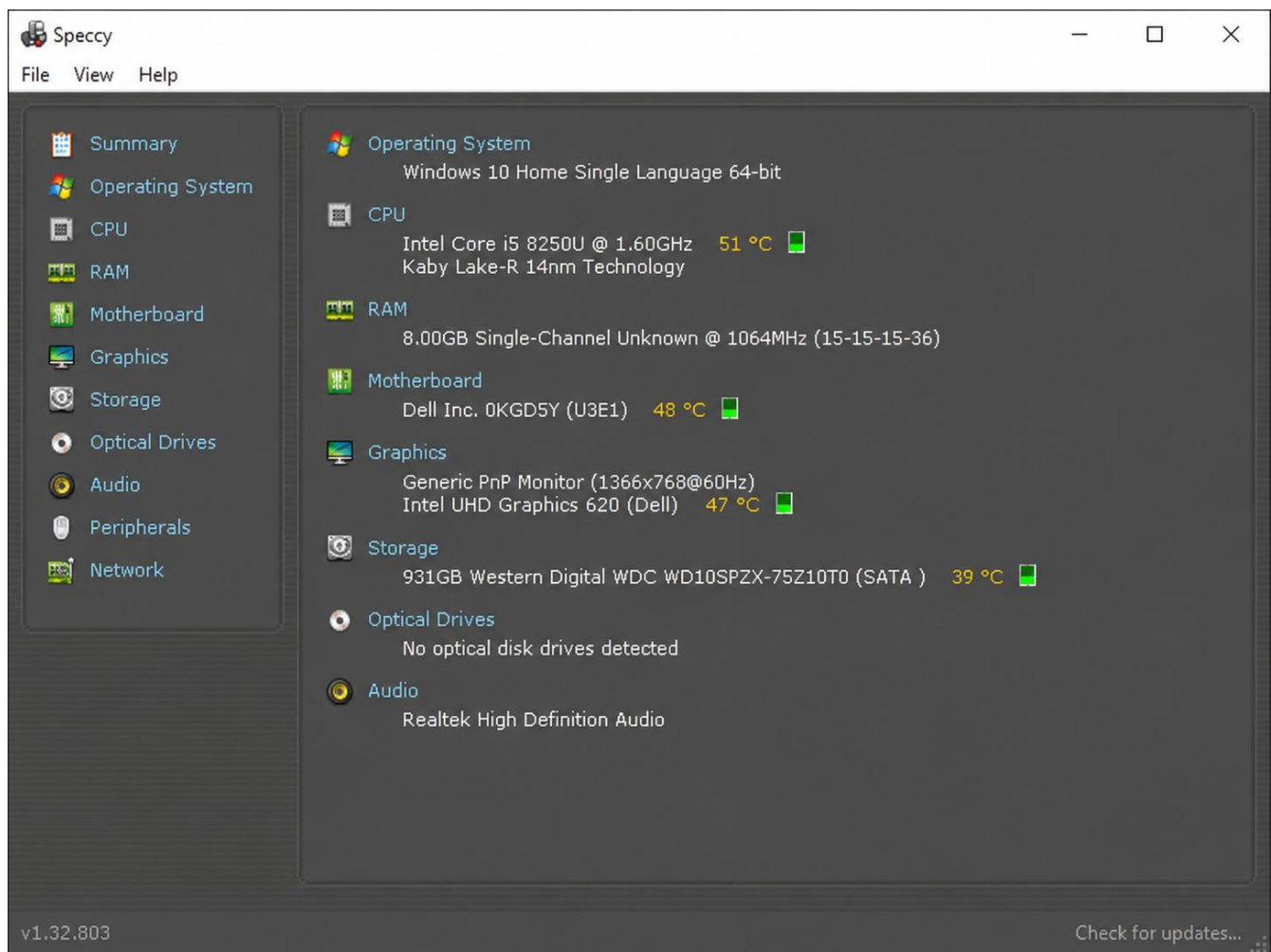
- Open Speccy from the Desktop shortcut or Start Menu. The software automatically scans the system and collects hardware information.



➤ Step 4: View the Main Dashboard:

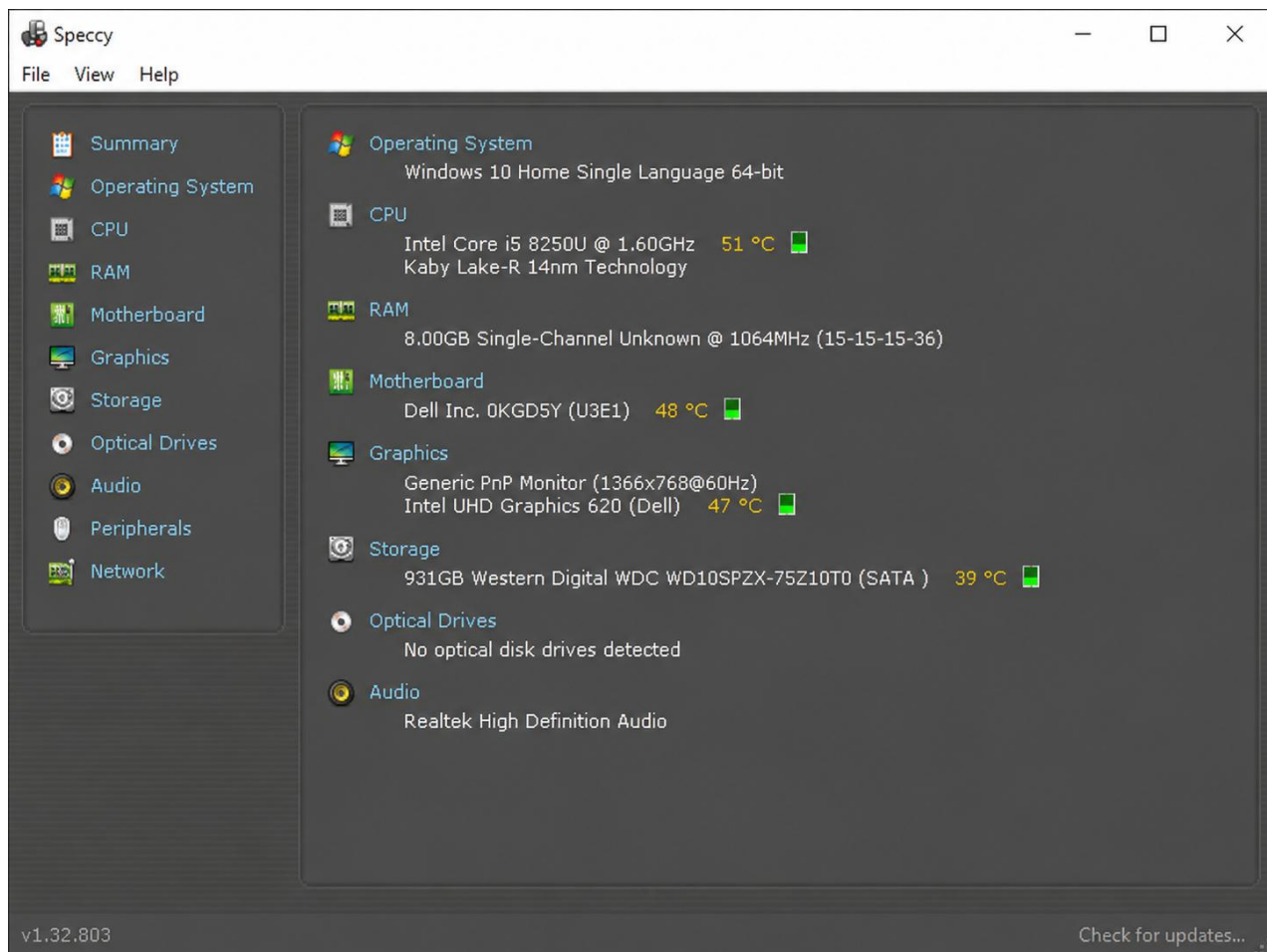
- After the scan is complete, the Summary dashboard displays:
- Operating System
- Processor (CPU)

- Motherboard
- RAM (Memory)
- Graphics Card (GPU)
- Storage Devices (HDD/SSD)
- System Temperature



➤ Step 5: Capture the Screenshot:

- Take a screenshot of the Summary dashboard.



- ❖ **4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, what it checks, Example Problem Detected) and fill in at least one example for each tool.**



Tool	What it Checks	Example (Problem Detected)
Multimeter	Measures voltage, current, resistance, and continuity in electrical circuits.	Detects a faulty SMPS, dead laptop charger, or damaged charging cable with no power output.
POST Card	Displays BIOS POST (Power-On Self-Test) codes during the computer startup process.	Identifies motherboard, RAM, CPU, or BIOS errors when the computer fails to boot.
Software Diagnostics (Speccy / HWMonitor / CrystalDiskInfo)	Monitors hardware information, temperatures, storage health, memory, and overall system performance.	Detects overheating, failing hard drives/SSDs, low memory, or abnormal CPU temperatures.